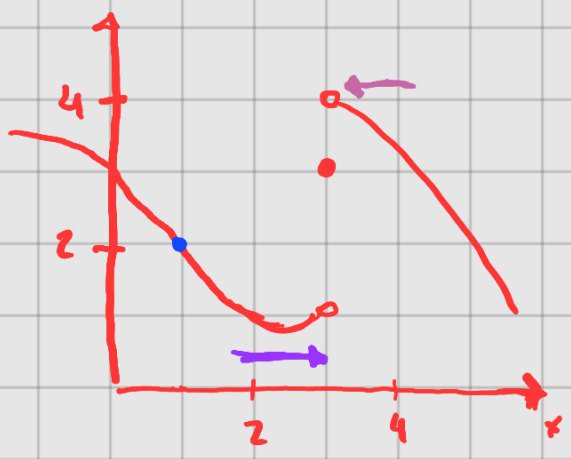
Calculo 07 e 08

28 -



1. $\lim_{x \to 1} f(x) = 2$

2. $\lim_{x \to 3^-} f(x) = 1$

3. $\lim_{x \to 3^+} f(x) = 4$

4. $\lim_{x \to 3} f(x) =$ Já que $\lim_{x \to 3^+} \neq \lim_{x \to 3^-}$ são diferentes, logo não existe

5. $f(3) = 4$

29 -

1. $\lim_{t \to 0^-} g(t) = -1$

2. $\lim_{t \to 0^+} g(t) = -2$

3. $\lim_{t \to 0} g(t) =$ Como $\lim_{t \to 0^+} \neq \lim_{t \to 0^-}$, então não existe

4. $\lim_{t \to 2^-} g(t) = 2$

3. $\lim_{t \to 2^+} g(t) = 0$

6. $\lim_{t \to 2} g(t) =$ Como $\lim_{t \to 0^-} \neq \lim_{t \to 0^+}$, então não existe

7. $g(2) = 1$

8. $g(0) = -1$

9. $\lim_{t \to 4} g(t) = 3$

30-

$$f(x)\begin{cases} 1+x, & \text{se } x<-1 \\ x^2, & \text{se } -1\leq x<1 \\ 2-x, & \text{se } x\geq 1 \end{cases}$$

$\lim\limits_{x\to -1^-} = -1$ $\qquad$ $\lim\limits_{x\to -1^+} = 1$

$\lim\limits_{x\to -1} =$ não existe já que $\lim\limits_{x\to -1^-} \neq \lim\limits_{x\to -1^+}$

$\lim\limits_{x\to 1^+} = 1$ $\qquad$ $\lim\limits_{x\to 1^-} = 1$

$\lim\limits_{x\to 1} = 1$

31-

a) $\lim\limits_{x\to -2} \dfrac{x^2+2x}{x+2} = \dfrac{\cancel{(x+2)}\cdot x}{\cancel{(x+2)}} = x = -2$

b) $\lim\limits_{x\to -1} \dfrac{3x^2+5x+2}{x+1} = \dfrac{(3x+2)\cdot\cancel{(x+1)}}{\cancel{(x+1)}} = 3x+2 = -1$

c) $\lim\limits_{x\to 2} \dfrac{2x^3-5x^2-4}{x^2-2x} = \dfrac{(2x+1)(x^2-2x)}{(x^2-2x)} = 2x+1 = 5$

32-

$$\lim_{x\to a} f(x) = 2 \; ; \; \lim_{x\to a} g(x) = -4 \; ; \; \lim_{x\to a} h(x) = 0$$

1. $\lim_{x\to a} (f(x) + 2g(x)) = -6$

2. $\lim_{x\to a} (h(x) - 3g(x) + 1) = 14$

3. $\lim_{x\to a} (2f(x) \cdot g(x)) = -16$

4. $\lim_{x\to a} \left(\sqrt[3]{6 + f(x)} \right) = \sqrt[3]{8} = 2$

5. $\lim_{x\to a} \left(f(x) + h(x) + 2g(x) + \frac{x^2 - a^2}{x - a} \right)$

$\lim_{x\to a} (2 \cdot 0 + (2 \cdot -4) + x + a)$

$-8 + a + a = -8 + 2a$

33-

1. $\lim_{x\to 2} (x^2 + 2x - 1) = 2^2 + 2 \cdot 2 - 1 = 7$

2. $\lim_{x\to 3} (2x^2 - 4x + 5) = 2 \cdot 3^2 - 12 + 5 = 11$

3. $\lim_{x\to -2} \frac{4x - 4}{5x - 1} = \frac{4 \cdot -2 - 4}{5 \cdot -2 - 1} = \frac{12}{11}$

4. $\lim_{x\to -1} \frac{2x - 1}{x^2 - 3x + 4} = \frac{(2 \cdot -1) - 1}{(-1)^2 - (3 \cdot -1) + 4} = \frac{-3}{8}$

5. $\lim_{r\to 1} \sqrt{\left(\frac{8r + 1}{r + 3}\right)} = \sqrt{\frac{8 + 1}{1 + 3}} = \frac{\sqrt{9}}{\sqrt{4}} = \frac{3}{2}$

6. $\lim_{x\to -1} (x^3 + x^2 + x)^{101} = ((-1)^3 + (-1)^2 + (-1))^{101} = (-1 + 1 - 1)^{101} = (-1)^{101} = -1$

7. $\lim_{x\to 4} \sqrt[3]{\frac{x^2 - 3x + 4}{2x^2 - x - 1}} = \sqrt[3]{\frac{4^2 - 3 \cdot 4 + 4}{2 \cdot 4^2 - 4 - 1}} = \sqrt[3]{\frac{8}{27}} = \frac{2}{3}$

8. $\lim_{x\to 7} \frac{x^2 - 49}{x - 7} = \frac{\cancel{(x - 7)}(x + 7)}{\cancel{(x - 7)}} = x + 7 = 7 + 7 = 14$

9 $\lim_{x \to 4} \left(\frac{3x^2 - 8x - 16}{2x^2 - 9x + 4} \right) = \frac{(3x+4)(x-4)}{(2x-1)(x-4)} = \frac{3x+4}{2x-1} = \frac{12+4}{8-1} = \frac{16}{7}$

10. $\lim_{x \to -2} \left(\frac{x^3 + 8}{x+2} \right) = x^2 - 2x + 4 = (-2)^2 - 2 \cdot -2 + 4 = 4$

11. $\lim_{x \to 4} \left(\frac{4 - x}{2 - \sqrt{x}} \right) = \frac{(2 - \sqrt{x}) \cdot (2 + \sqrt{x})}{2 - \sqrt{x}} = 2 + \sqrt{4} = 4$

12. $\lim_{x \to -3} \sqrt{\frac{x^2 - 9}{2x^2 + 7x + 3}} = \frac{\cancel{(x+3)} \cdot (x-3)}{\cancel{(x+3)} \cdot (2x+1)} = \frac{x-3}{2x+1} = \frac{-6}{-5} = \frac{6}{5}$

13. $\lim_{x \to 0} \frac{\sqrt{x+9} - 3}{x} = \frac{\sqrt{x+9} - 3}{x} \cdot \frac{\sqrt{x+9} + 3}{\sqrt{x+9} + 3} = \frac{(\sqrt{x+9})^2 \cdot (-3)^2}{x(\sqrt{x+9} + 3)}$

$\frac{\cancel{x}}{\cancel{x} \cdot \sqrt{x+9} + 3} = \frac{1}{\sqrt{x+9} + 3} = \frac{1}{6}$

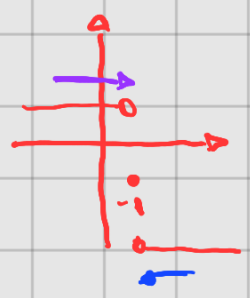
34.

$f(x) \begin{cases} 2, \text{ se } x < 1 \\ -1, \text{ se } x = 1 \\ -3, \text{ se } 1 < x \end{cases}$

a) $\lim_{x \to 1^+} = -3$

b) $\lim_{x \to 1^-} = 2$

c) $\lim_{x \to 1} =$ não existe



$f(x) \begin{cases} 2x+3, \text{ se } x < 1 \\ 4, \text{ se } x = 1 \\ x^2+2, \text{ se } 1 < x \end{cases}$

a) $\lim_{x \to 1^+} = 3$

b) $\lim_{x \to 1^-} = 5$

c) $\lim_{x \to 1} =$ não existe

$$f(x)\begin{cases} x^2, & \text{se } x \leq 2 \\ 8-2x, & \text{se } 2 < x \end{cases}$$

a) $\lim_{x \to 2^+} f(x) = 4$

b) $\lim_{x \to 2^-} f(x) = 4$

c) $\lim_{x \to 2} f(x) = 4$

$f(x) = 3 + |2x-4|$

a) $\lim_{x \to 2^+} = 3$

b) $\lim_{x \to 2^-} = 3$

c) $\lim_{x \to 2} = 3$

35.

$$f(x)\begin{cases} \dfrac{x^2-9}{x+3}, & \text{se } x \neq 3 \\ k, & \text{se } x = -3 \end{cases}$$

$$\frac{x^2-9}{x+3} = \frac{\cancel{(x+3)} \cdot (x-3)}{\cancel{(x+3)}}$$

$= x-3 \qquad x = -3$

$-3-3 = -6$ //

$k = -6$